

Solve multi-step multiplication and division problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A warehouse has 36 pallets with 24 boxes each. It shares all boxes equally among 18 shops. Boxes per shop? Copy or complete the first step.

2. A printer makes 1,250 leaflets per hour for 28 hours and packs them in bundles of 50. How many bundles?

3. A farmer packs 9,216 apples in trays of 24, then places 16 trays on each trolley. How many trolleys?

4. Miro says: Miro rounds an intermediate exact answer before completing the second step. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A warehouse has 36 pallets with 24 boxes each. It shares all boxes equally among 18 shops. Boxes per shop? Copy or complete the first step.

Answer 48 boxes.

2. A printer makes 1,250 leaflets per hour for 28 hours and packs them in bundles of 50. How many bundles?

Answer 700 bundles.

3. A farmer packs 9,216 apples in trays of 24, then places 16 trays on each trolley. How many trolleys?

Answer 24 trolleys.

4. Miro says: Miro rounds an intermediate exact answer before completing the second step. Is this correct? Explain.

Answer Keep exact sub-totals unless the context or question asks for rounding.

Solve multi-step multiplication and division problems

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A warehouse has 36 pallets with 24 boxes each. It shares all boxes equally among 18 shops. Boxes per shop?

6. Check this result using an inverse, estimate or known fact: A farmer packs 9,216 apples in trays of 24, then places 16 trays on each trolley. How many trolleys?

2. A printer makes 1,250 leaflets per hour for 28 hours and packs them in bundles of 50. How many bundles?

7. A result is 315 after a two-step multiply-then-divide calculation. Create a valid problem.

3. A farmer packs 9,216 apples in trays of 24, then places 16 trays on each trolley. How many trolleys?

8. Identify and correct this misconception: Miro rounds an intermediate exact answer before completing the second step.

4. Solve this independently and show every stage: A warehouse has 36 pallets with 24 boxes each. It shares all boxes equally among 18 shops. Boxes per shop?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A printer makes 1,250 leaflets per hour for 28 hours and packs them in bundles of 50. How many bundles?

10. Write and solve a similar question to: A farmer packs 9,216 apples in trays of 24, then places 16 trays on each trolley. How many trolleys?

Teacher answers and checking prompts

1. A warehouse has 36 pallets with 24 boxes each. It shares all boxes equally among 18 shops. Boxes per shop?
Answer 48 boxes.
2. A printer makes 1,250 leaflets per hour for 28 hours and packs them in bundles of 50. How many bundles?
Answer 700 bundles.
3. A farmer packs 9,216 apples in trays of 24, then places 16 trays on each trolley. How many trolleys?
Answer 24 trolleys.
4. Solve this independently and show every stage: A warehouse has 36 pallets with 24 boxes each. It shares all boxes equally among 18 shops. Boxes per shop?
Answer 48 boxes.
5. Use a second method or representation for: A printer makes 1,250 leaflets per hour for 28 hours and packs them in bundles of 50. How many bundles?
Answer Expected result: 700 bundles. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A farmer packs 9,216 apples in trays of 24, then places 16 trays on each trolley. How many trolleys?
Answer Expected result: 24 trolleys. A valid check must be shown.
7. A result is 315 after a two-step multiply-then-divide calculation. Create a valid problem.
Answer Answers vary, for example $45 \times 21 \div 3 = 315$.
8. Identify and correct this misconception: Miro rounds an intermediate exact answer before completing the second step.
Answer Keep exact sub-totals unless the context or question asks for rounding.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A farmer packs 9,216 apples in trays of 24, then places 16 trays on each trolley. How many trolleys?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve multi-step multiplication and division problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. A result is 315 after a two-step multiply-then-divide calculation. Create a valid problem.

6. Create a new example that tests the same idea as: A farmer packs 9,216 apples in trays of 24, then places 16 trays on each trolley. How many trolleys?

2. Prove the answer to this question, rather than only calculating it: A warehouse has 36 pallets with 24 boxes each. It shares all boxes equally among 18 shops. Boxes per shop?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A printer makes 1,250 leaflets per hour for 28 hours and packs them in bundles of 50. How many bundles?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 24 trolleys.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro rounds an intermediate exact answer before completing the second step.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. A result is 315 after a two-step multiply-then-divide calculation. Create a valid problem.
Answer Answers vary, for example $45 \times 21 \div 3 = 315$.
2. Prove the answer to this question, rather than only calculating it: A warehouse has 36 pallets with 24 boxes each. It shares all boxes equally among 18 shops. Boxes per shop?
Answer Expected result: 48 boxes. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A printer makes 1,250 leaflets per hour for 28 hours and packs them in bundles of 50. How many bundles?
Answer Expected result: 700 bundles. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 24 trolleys.
Answer One valid reconstruction is: A farmer packs 9,216 apples in trays of 24, then places 16 trays on each trolley. How many trolleys? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro rounds an intermediate exact answer before completing the second step.
Answer Keep exact sub-totals unless the context or question asks for rounding.
6. Create a new example that tests the same idea as: A farmer packs 9,216 apples in trays of 24, then places 16 trays on each trolley. How many trolleys?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.